

SERVERLESS IMAGE PROCESSING SYSTEM

1. Project Objective

- The objective is to automatically handle an image when a user uploads it to Amazon S3.

User uploads image:

- S3 – input/ → S3 Event Trigger → AWS Lambda → S3 - processed/

We use only:

*Amazon S3 – store images

*AWS Lambda – automatically process/copy the image

*IAM – provide permissions to Lambda

*CloudWatch – view Lambda logs

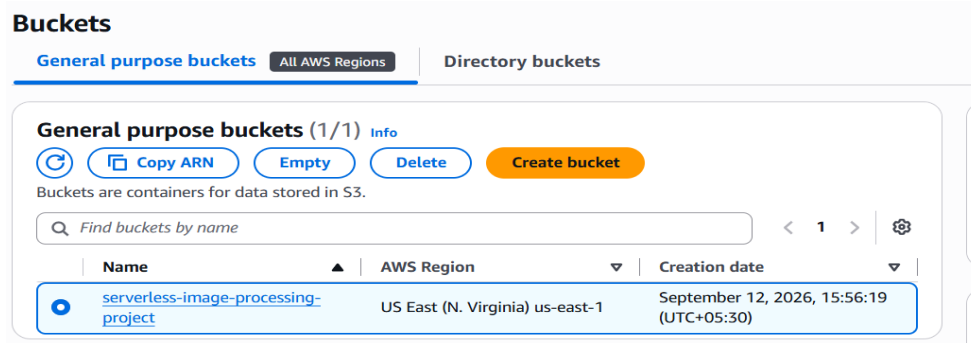
2. Create an S3 Bucket

> Go to:

> AWS Console → S3 → Create bucket

> Enter a unique bucket name, for example:

serverless-image-processing-project



General configuration

AWS Region
US East (N. Virginia) us-east-1

Bucket type [Info](#)

☒ **General purpose**
Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.

☐ **Directory**
Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

Bucket namespace
Choose the namespace where you want to create your bucket. [Learn more](#)

☒ **Global namespace**
By default, S3 creates general purpose buckets in the global namespace.

☐ **Account Regional namespace (recommended)**
General purpose buckets created in your account Regional namespace are unique to your account. These buckets can never be created by another AWS account.

Bucket name [Info](#)
serverless-image-processing-project

Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). [Learn more](#)

Copy settings from existing bucket - optional
Only the bucket settings in the following configuration are copied.

[Choose bucket](#)

- **Choose your required AWS Region.**
- **Keep: Block all public access → Enabled**
- **Keep other settings as default.**
- **Click: Create bucket**

3. Create Two Folders

- > **Open your bucket.**
- > **Create these two folders**

→ **input/**

→ **processed/**

serverless-image-processing-project [Info](#)

[Objects](#) | [Metadata](#) | [Properties](#) | [Permissions](#) | [Metrics](#) | [Management](#) | [File systems](#) | [Access Points](#)

Objects (2/2)

[Refresh](#) [Copy S3 URI](#) [Copy URL](#) [Download](#) [Open](#) [Delete](#) [Actions](#) [Create folder](#) [Upload](#)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

☒	Name	Type	Last modified	Size	Storage class
☒	input/	Folder	-	-	-
☒	processed/	Folder	-	-	-

➤ Purpose

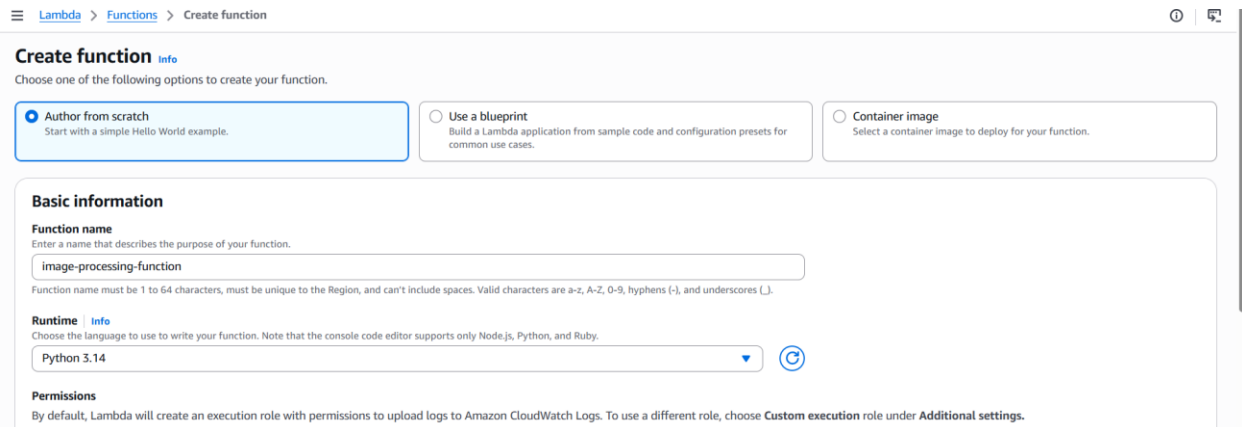
***input/ → original images are uploaded here.**

***processed/ → Lambda stores the processed output here.**

4. Create the Lambda Function

> **Go to:**

> **AWS Console → Lambda → Create function**



The screenshot shows the 'Create function' page in the AWS Lambda console. At the top, there's a breadcrumb trail: 'Lambda > Functions > Create function'. Below this, the title 'Create function' is followed by an 'info' icon. A subtitle reads: 'Choose one of the following options to create your function.' There are three radio button options: 'Author from scratch' (selected), 'Use a blueprint', and 'Container image'. The 'Author from scratch' option has a subtext: 'Start with a simple Hello World example.' Below these options is a section titled 'Basic information'. It contains a 'Function name' field with the value 'image-processing-function' and a subtext: 'Enter a name that describes the purpose of your function.' Below the name field is a note: 'Function name must be 1 to 64 characters, must be unique to the Region, and can't include spaces. Valid characters are a-z, A-Z, 0-9, hyphens (-), and underscores (_).' There is also a 'Runtime' dropdown menu set to 'Python 3.14' with a subtext: 'Choose the language to use to write your function. Note that the console code editor supports only Node.js, Python, and Ruby.' At the bottom, there is a 'Permissions' section with a subtext: 'By default, Lambda will create an execution role with permissions to upload logs to Amazon CloudWatch Logs. To use a different role, choose Custom execution role under Additional settings.'

***Select: Author from scratch**

***Function name: image-processing-function**

***Runtime: Python 3.14**

***Click: Create function**

5. Give Lambda S3 Permission

6. Configure S3 Trigger

> **Go to your Lambda function.**

> **Click: Add trigger**

***Select: S3**

***Choose your S3 bucket.**

Add trigger

Trigger configuration [Info](#)



aws asynchronous storage

Bucket

Choose or enter the ARN of an S3 bucket that serves as the event source. The bucket must be in the same region as the function.



Bucket region: us-east-1

Event types

Select the events that you want to have trigger the Lambda function. You can optionally set up a prefix or suffix for an event. However, for each bucket, individual events cannot have multiple configurations with the same object key.

All object create events

***Event type: All object create events**

***For Prefix, enter: input/**

7. Test the Project

> **Go to:**

> **S3 → Your Bucket → input/**

> **Click: Upload**

> **Select an image, for example:**

> **photo.jpg..... Upload it.**

Upload [Info](#)

Add the files and folders you want to upload to S3. To upload a file larger than 160GB, use the AWS CLI, AWS SDKs or Amazon S3 REST API. [Learn more](#)

Drag and drop files and folders you want to upload here, or choose **Add files** or **Add folder**.

Files and folders (1 total, 58.0 KB)

All files and folders in this table will be uploaded.

☐	Name	Folder	Type	Size
☐	japan desktop wallpaper.jpg	-	image/jpeg	58.0 KB

Destination [Info](#)

Destination

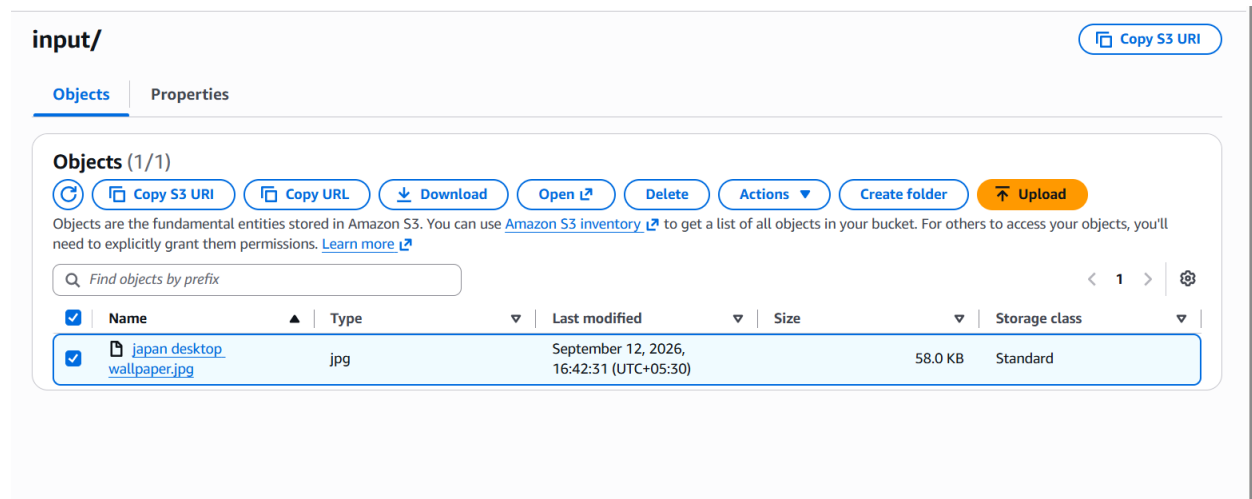
<s3://serverless-image-processing-project/input/>

► Destination details

> You should have:

*input/

└── photo.jpg



input/ [Copy S3 URI](#)

Objects Properties

Objects (1/1)

[Refresh](#) [Copy S3 URI](#) [Copy URL](#) [Download](#) [Open](#) [Delete](#) [Actions](#) [Create folder](#) [Upload](#)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

☒	Name	Type	Last modified	Size	Storage class
☒	japan desktop wallpaper.jpg	jpg	September 12, 2026, 16:42:31 (UTC+05:30)	58.0 KB	Standard

8. Check the Processed Folder

> Go to:

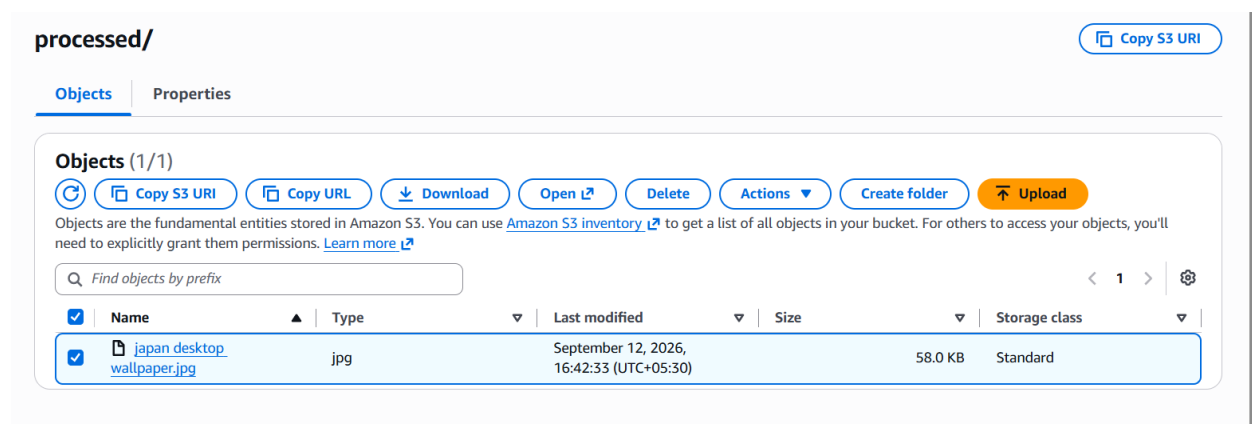
>S3 → Your Bucket → processed/

> You should see:

*processed/

└── photo.jpg

****If the file appears, your project is working successfully. ****



processed/ [Copy S3 URI](#)

Objects Properties

Objects (1/1)

[Refresh](#) [Copy S3 URI](#) [Copy URL](#) [Download](#) [Open](#) [Delete](#) [Actions](#) [Create folder](#) [Upload](#)

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

☒	Name	Type	Last modified	Size	Storage class
☒	japan desktop wallpaper.jpg	jpg	September 12, 2026, 16:42:33 (UTC+05:30)	58.0 KB	Standard

9. Check CloudWatch Logs

>Go to:

> **Lambda** → **Your Function** → **Monitor** → **View CloudWatch logs**

Log events

Actions

Start tailing

Create metric filter

You can use the filter bar below to search for and match terms, phrases, or values in your log events. [Learn more about filter patterns](#)

Q

Filter events - press enter to search

Clear

1m

30m

1h

12h

Custom

UTC timezone

Display

Timestamp	Message
	No older events at this moment. Retry
2026-09-12T11:12:31.418Z	INIT_START Runtime Version: python:3.14.mainline.v66 Runtime Version ARN: arn:aws:lambda:us-east-1::runtime:560c77f6cdda2265a11ed...
2026-09-12T11:12:31.985Z	START RequestId: 831b18a3-a4b1-4063-8703-85b8181ed9cd Version: \$LATEST
2026-09-12T11:12:32.294Z	Image processed successfully
2026-09-12T11:12:32.294Z	Original: input/japan desktop wallpaper.jpg
2026-09-12T11:12:32.294Z	Processed: processed/japan desktop wallpaper.jpg
2026-09-12T11:12:32.332Z	END RequestId: 831b18a3-a4b1-4063-8703-85b8181ed9cd
2026-09-12T11:12:32.332Z	REPORT RequestId: 831b18a3-a4b1-4063-8703-85b8181ed9cd Duration: 345.50 ms Billed Duration: 909 ms Memory Size: 128 MB Max Memory...

>You should see messages such as:

***Image processed successfully**

***Original: input/photo.jpg**

***Processed: processed/photo.jpg**